THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Tangents PA and PB are drawn from a point P, 50 cm from the centre O of a circle of radius 14 cm. The length of each tangent is 48 cm. The length of the chord AB is: 1

(a) 96 cm **(b)** 26.88 cm **(c)** 28 cm **(d)** 13.44 cm

2. A chord of a circle of radius 10.5 cm subtends an angle of 120° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) The area of the minor segment is: 1

(a) 47.68 cm² **(b)** 163.18 cm² **(c)** 67.82 cm² **(d)** 115.50 cm²

3. If $\sin\theta = \frac{12}{13}$, the value of $2\sin\theta\cos\theta$ is: 1

(a) $\frac{10}{13}$ **(b)** $\frac{24}{13}$ **(c)** $\frac{60}{169}$ **(d)** $\frac{120}{169}$

4. A flagstaff stands on the top of a building 12 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 30° and 60° respectively. The length of the flagstaff is: 1

(a) 24 m **(b)** 36 m **(c)** 48 m **(d)** $24\sqrt{3}$ m

5. PA and PB are tangents from P to a circle, each of length 11 cm. A third tangent touches the circle at E on the minor arc AB and meets PA at C and PB at D. The perimeter of $\triangle PCD$ is: 1

(a) 22 cm **(b)** 13 cm **(c)** 11 cm **(d)** 33 cm

6. A brooch is a circular piece of silver wire of diameter 49 mm. The wire is also used for 7 equal diameters that divide the brooch into 14 sectors. The total length of wire required (in mm) is: 1

(a) 497 mm **(b)** 154 mm **(c)** 1886.5 mm **(d)** 325.5 mm

7. If $\sec\theta + \tan\theta = 5$, then $\sec\theta - \tan\theta =$ 1

(a) 5 **(b)** $\frac{1}{5}$ **(c)** 2 **(d)** 1

8. A car moves at a uniform speed towards the foot of a vertical tower. The angle of elevation of the top of the tower changes from 30° to 60° in 8 minutes. The further time the car takes to reach the foot of the tower is: 1

(a) $\frac{8}{3}$ minutes **(b)** 4 minutes **(c)** 16 minutes **(d)** 8 minutes

9. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (12x + 7)$ cm and $QB = (5x + 28)$ cm. The length of QA is: 1

(a) 3 cm **(b)** 43 cm **(c)** 45 cm **(d)** 64 cm

10. A horse is tied at the midpoint of one side of a large square field with a rope 42 m long. The area of the field it can graze is: 1

(a) 2772 m² **(b)** 132 m² **(c)** 66 m² **(d)** 5544 m²

11. $\dfrac{\sin^2 40^\circ + \sin^2 50^\circ}{\cos^2 12^\circ + \cos^2 78^\circ} =$ 1

(a) 2 **(b)** 1 **(c)** 0 **(d)** $\frac{1}{2}$

12. From the top of a tower $9\sqrt{3}$ m high, the angles of depression of two objects on level ground, on the same side of the tower and in line with its foot, are 30° and 60°. The distance between the two objects is: 1

(a) $18\sqrt{3}$ m **(b)** 36 m **(c)** 9 m **(d)** 18 m

13. **Assertion (A):** Exactly two tangents can be drawn to a circle from a point outside it. 1

Reason (R): A tangent to a circle intersects the circle at two points.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** The length of the arc of a sector of radius 21 cm and angle 60° is 22 cm (use $\pi = \frac{22}{7}$). 1

Reason (R): The length of the arc of a sector of angle θ is $\dfrac{\theta}{360^\circ} \times \pi r^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** $\sin\theta$ can be equal to 1.5 for some acute angle θ. 1

Reason (R): In a right triangle the hypotenuse is the longest side, so $\sin\theta = \dfrac{\text{perpendicular}}{\text{hypotenuse}} \leq 1$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Two concentric circles have radii 90 cm and 72 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. 2

17. A circular park of radius 35 m is surrounded by a path 3.5 m wide. Find the area of the path. 2

18. If $\sin\theta = \frac{12}{13}$, find the value of $\cos^2\theta - \sin^2\theta$. 2

19. The angle of elevation of the top of a vertical tower from a point on level ground is 45°. After walking 12 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower. 2

20. Tangents PA and PB are drawn from a point P, 75 cm from the centre O of a circle of radius 72 cm. The length of each tangent is 21 cm. Find the length of the chord AB. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Tangents PA and PB are drawn from a point P, 50 cm from the centre O of a circle of radius 14 cm. The length of each tangent is 48 cm. Find the length of the chord AB. 3

22. A sector of a circle of radius 42 cm has central angle 80°. Find the area of the sector. 3

23. If $\sin\theta + \cos\theta = \frac{13}{10}$, find $\sin\theta\cos\theta$. 3

24. A flagstaff stands on the top of a building 16 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 45° and 60° respectively. Find the length of the flagstaff. 3

OR

In $\triangle ABC$, right-angled at B, $AB = 40$ cm and $BC = 75$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

25. A quadrant of a circle of radius 70 cm is cut from a square of side 70 cm, with the centre of the quadrant at one corner of the square. Find the area of the remaining part of the square. 3

OR

If $\tan\theta + \cot\theta = 4$, find $\tan^2\theta + \cot^2\theta$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** A circular park has centre O and radius 6 m. A gate P is at a distance of 10 m from O, outside the park. Two straight paths PA and PB touch the park at the points A and B. 5

(i) What is the measure of the angle between OA and the path PA?

(ii) Find the length of the path PA.

(iii) Write the length of the path PB.

(iv) Find the area of the quadrilateral OAPB.

27. Prove that in a right triangle ABC with $\angle B = 90^\circ$, the radius of the incircle is $r = \frac{AB + BC - AC}{2}$. 5

OR

A chord of a circle of radius 42 cm subtends an angle of 120° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. 5

END OF QUESTION PAPER — **Total: 50 Marks**